



AQ1.5: Activity Questions 5 - Not Graded



This assignment will not be graded and is only for practice.

5.1 Level 1:

1 point

Choose the set of correct options

☐

If AA is a real $3 \times 3 \times 3$ matrix, then $\det(A) = \det(A^T)\det(A) = \det(AT)$.

☐

If $A = [a_{ij}]$ $A = [a_{ij}]$ is a real $4 \times 4 \times 4$ matrix, then the order of the sub matrix obtained by deleting the ii -th row and jj -th column of AA is $4 \times 4 \times 4$.

☐

If $A = [a_{ij}]$ $A = [a_{ij}]$ is a real $4 \times 4 \times 4$ matrix, then the order of the sub matrix obtained by deleting the ii -th row and jj -th column of AA is $3 \times 3 \times 3$.

☐

If AA is a real $3 \times 3 \times 3$ matrix, then the orders of all possible square sub matrices of AA are 1×1 , 2×2 , and 3×3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If AA is a real $3 \times 3 \times 3$ matrix, then $\det(A) = \det(A^T)\det(A) = \det(AT)$.

If $A = [a_{ij}]$ $A = [a_{ij}]$ is a real $4 \times 4 \times 4$ matrix, then the order of the sub matrix obtained by deleting the ii -th row and jj -th column of AA is $3 \times 3 \times 3$.

If AA is a real $3 \times 3 \times 3$ matrix, then the orders of all possible square sub matrices of AA are 1×1 , 2×2 , and 3×3 .

1 point

Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix}$ be a $3 \times 3 \times 3$ matrix. Which of the following is(are) correct?

[Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

☐

$\det(A) = 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) - 2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 3 \\ 2 & 2 \end{bmatrix}\right)$ $\det(A) = 3 \times \det([3121]) - 2 \times \det([2221]) + 2 \times \det([2231])$

☐

$\det(A) = 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right)$ $\det(A) = 3 \times \det([3121]) + 2 \times \det([2122]) + 2 \times \det([2231])$

☐

$\det(A) = -2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) + 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) - 2 \times \det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right)$ $\det(A) = -2 \times \det([2121]) + 3 \times \det([3221]) - 2 \times \det([3221])$

☐

$\det(A) = 2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) - 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right)$ $\det(A) = 2 \times \det([2121]) - 3 \times \det([3221]) + 2 \times \det([2231])$



$$]) + 2 \times \det([3221])$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(A) = 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\right) - 2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right) \det(A) = 3 \times \det([3121]) - 2 \times \det([2221]) + 2 \times \det([2231])$$

$$\det(A) = 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}\right) + 2 \times \det\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}\right) \det(A) = 3 \times \det([3121]) + 2 \times \det([2122]) + 2 \times \det([2231])$$

$$\det(A) = -2 \times \det\left(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}\right) + 3 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) - 2 \times \det\left(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) \det(A) = -2 \times \det([2121]) + 3 \times \det([3221]) - 2 \times \det([3221])$$

1 point

If all the elements of a $3 \times 3 \times 3$ real matrix AA are the same, then which of the following is (are) correct?

☐

Determinant of matrix AA is 00.

☐

Determinant of matrix AA cannot be determined from the given information.

☐

Determinant of matrix AA will be the sum of the elements of a row.

☐

Determinant of matrix $A + A^T A + AT$ is 00, where $A^T AT$ denotes the transpose of AA .

☐

Determinant of matrix $A + A^T A + AT$ cannot be determined from the given information, where $A^T AT$ denotes the transpose of AA .

No, the answer is incorrect.

Score: 0

Accepted Answers:

Determinant of matrix AA is 00.

Determinant of matrix $A + A^T A + AT$ is 00, where $A^T AT$ denotes the transpose of AA .

1 point

Choose the set of correct options

☐

If two rows are the same in a $3 \times 3 \times 3$ real matrix, then the determinant of that matrix is zero.

☐

If two columns are the same in a $3 \times 3 \times 3$ real matrix, then the determinant of that matrix is zero.

☐

If one row is a non-zero multiple of another row in a $3 \times 3 \times 3$ real matrix, then the determinant of that matrix is not zero.

☐

If one column is a non-zero multiple of another column in a $3 \times 3 \times 3$ real matrix, then the determinant of that matrix is zero.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If two rows are the same in a $3 \times 3 \times 3$ real matrix, then the determinant of that matrix is zero.

If two columns are the same in a $3 \times 3 \times 3$ real matrix, then the determinant of that matrix is zero.

If one column is a non-zero multiple of another column in a $3 \times 3 \times 3$ real matrix, then the determinant

of that matrix is zero.

Let $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ be a matrix of order 4×4 . Find $\det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

5.2 Level 2:

1 point

Let A, B, A, B , and C be 3×3 real matrices. Which of the following options is (are) correct?

☐

$$\det(ABC) = \det(A)\det(B)\det(C) \quad \det(ABC) = \det(A)\det(B)\det(C)$$

☐

$$\det(A^3) = \det(A)^3 \quad \det(A^3) = \det(A)^3$$

☐

$$\det(A + B + C) = \det(A) + \det(B) + \det(C) \quad \det(A + B + C) = \det(A) + \det(B) + \det(C)$$

☐

$$\det(AB^T) = \det(A)\det(B) \quad \det(AB^T) = \det(A)\det(B), \text{ where } B^T \text{ denotes the transpose of } B.$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(ABC) = \det(A)\det(B)\det(C) \quad \det(ABC) = \det(A)\det(B)\det(C)$$

$$\det(A^3) = \det(A)^3 \quad \det(A^3) = \det(A)^3$$

$$\det(AB^T) = \det(A)\det(B) \quad \det(AB^T) = \det(A)\det(B), \text{ where } B^T \text{ denotes the transpose of } B.$$

1 point

Let $A = [\delta_{ij}]$ be a matrix of order 3×3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i > j \\ 1 & \text{if } i \leq j \end{cases}$

Choose the set of correct options

[Hint: Write down matrix A first.]

☐

A is an upper triangular matrix.

☐

A is a lower triangular matrix.

☐

$$\det(A) = 6 \quad \det(A) = 6$$

☐

$$\det(A) = 3 \quad \det(A) = 3$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

AA is an upper triangular matrix.

$$\det(A) = 6\det(A) = 6$$

1 point

Let $A = [\delta_{ij}]$ be a matrix of order 3×3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i < j \\ 2 & \text{if } i \geq j \end{cases}$

Choose the set of correct options

☐

AA is an upper triangular matrix.

☐

AA is a lower triangular matrix.

☐

$$\det(A) = 2\det(A) = 2$$

☐

$$\det(A) = 8\det(A) = 8$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

AA is a lower triangular matrix.

$$\det(A) = 8\det(A) = 8$$

Suppose $A = \begin{bmatrix} 4 & -13 \\ 2 & 0 \\ 3 & -20 \end{bmatrix}$ and C_{ij} is the (i, j) -th cofactor of the matrix AA .

Let BB be 3×3 matrix where (i, j) -th entry is equal to C_{ij} . Find $\det(B)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 49

1 point

Suppose AA and BB are two 3×3 matrices such that $\det(A) = 4$ and $B = 3A$. Find the value of $\sqrt[3]{\det(A^2 B)}$

.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 12

1 point

[Check Answers](#)

Your score is: 0/10